

Heartbeat Classifier for PPG Signals. Assignment Report

Applied AI in Biomedicine

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Introduction

Photoplethysmography (PPG) is a noninvasive optical method for measuring changes in tissue blood volume, widely used to monitor cardiovascular activity. In recent years, PPG has become one of the most popular methods of analysis in cardiology, pulmonology and general health monitoring medicine due to its simplicity and availability. Photoplethysmography can be obtained not only using standard medical equipment – a photoplethysmograph, but also using a smart watch [1].

Photoplethysmography measures changes in blood volume in vascular tissue using optical radiation that passes through peripheral tissues, where it is scattered and absorbed by various tissues, and then either transmitted through them or reflected from the surface. This attenuated light emission is detected by an optical sensor and recorded as a voltage signal known as a PPG [2]. The original PPG wave reflects changes in the attenuation of the incident optical radiation by various components within the tissue volume, as shown in fig.1.

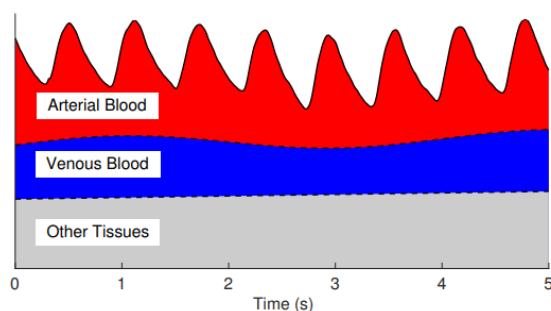


Figure 1. The simplified origins of the PPG signal: it is primarily influenced by changes in the volumes of arterial blood, venous blood and other tissues.

High frequency variations are caused by changes in arterial blood volume with each

heartbeat, and low frequency variations are associated with changes in other tissue components such as venous and capillary blood, bloodless tissue, etc. The attenuation of light in tissue can be described as a function of the optical path length and the attenuation coefficient of the medium, based on the modified Beer-Lambert law. [2]. In PPG signal analysis, three types of peaks could be defined: N, S and V. N-peaks (Normal peaks) occur during normal sinus rhythm with regular intervals between beats (NN interval). S-peaks (Supraventricular peaks) are associated with supraventricular arrhythmias, where the electrical activity originates from the atria or atrioventricular node, while there is no compensation interval, and the intervals between the peaks remain irregular but short. V-peaks (Ventricular peaks) occur in ventricular arrhythmias such as extrasystole and are characterized by a complete compensatory interval, when after the V-peak there is a prolonged interval until the next normal beat (VV interval).

Dataset and task setting

A dataset – 105 records including PPG signals, peaks and labels – was provided.

In this project, the main task was to create a heartbeat classifier based on PPG signals.

It was necessary to perform two consecutive classifications:

1. Determine normal (N) and abnormal (non-N) beats.
2. Classify peaks as N, S, V.

To correctly complete this task, it was necessary to preprocess the data, build model

structures and develop metrics to improve the accuracy of peak classification.

Data preprocessing

Before starting the work, the dataset was analyzed for missing data, as well as the distribution of peaks of different types. Basically, all peaks are classified as normal (N) – on average, there are more than 90% of them. This means that in the future we must be more careful when dividing the data into sets and balance the distribution of different types of peaks.

Resampling

After studying the dataset, it became clear that the signals contained in it have different frequencies – 128 Hz and 250 Hz. Therefore, the first step was to resample the signals to a frequency of 250 Hz.

Filtration

Next, the signals were filtered from excess noise.

A classic lowpass filter was used, which removed all noise below 5 Hz (the frequency was chosen in accordance with the literature [3]). During the work, experiments were also conducted with the Kalman filter, which is adaptive and changes the cutoff frequency to the most suitable at each step, which should be the most universal and reliable solution. However, due to the lack of RAM resources, as well as due to the very long signal processing when using such a filter, it was decided to use a regular lowpass.

Normalization & Label encoding

Initially, for convenience and simplification of the model, min-max normalization was used, which brought the peak amplitudes to values from 0 to 1. However, further experiments showed that such normalization has a negative effect on the ability of the model to recognize classes S and V, which is especially important for the second model. Therefore, after studying the influence of

different types of normalization, z-score was chosen.

The label values were converted to binary form to improve and facilitate the feeding of data into the model. For this, the OneHotEncoder function was used.

The example of signal after resampling, filtration and normalization is shown in fig.2.

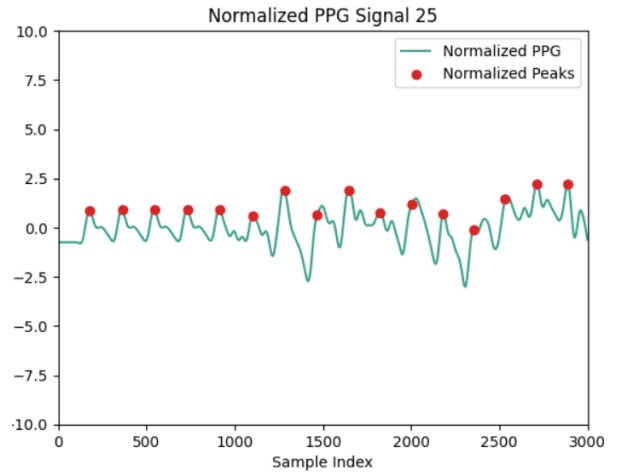


Figure 2. Normalized PPG signal with peaks

Split dataset to training, validation and test sets

After signal processing, the data was divided into training, validation, and test sets in a ratio of 75 to 20 to 5.

Segmentation

Since the classification of peaks is based on the intervals between them, it is important to preserve the dimensionality of these intervals when segmenting the data. Therefore, the following sections of the PPG signal were selected as a segment:

After dividing into sets, the signals were divided into segments containing one peak each, 400 in length. This was done as follows: From the original signal, the part between two peaks containing the central peak was extracted. Then the length of the resulting segment was adjusted - if the segment was longer than 400 pixels, it was cut off, if shorter, it was filled with zeros using zero padding. After that, the peak position was

recalculated relative to the beginning of the segment.

The example of segment is shown in fig. 3.

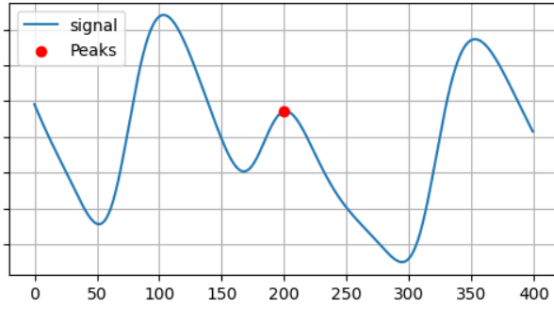


Figure 3. One segment of PPG signal (length = 700)

LST model for N vs non-N classification

To classify peaks into en and non-en, models of different types were tested, including SNN, LSTM, bidirectional LSTM, SNN-LSTM and GRU. The highest efficiency with a sufficiently fast training speed and acceptable RAM costs.

The final model was LSTM with a total number of parameters equal to 133,538.

It includes three LSTM layers, two fully connected layers, three normalization layers and 4 dropout layers.

To combat overfitting, the following regularization methods were used: recurrent dropout with the parameter 0.2 disables some neurons in recurrent connections, and the L2 kernel regularizer limits the growth of weights.

The output layer of the model consists of 2 neurons, which corresponds to binary classification. It uses the softmax activation function, which transforms the outputs into probabilities for each of the two classes.

To optimize the model, Adam is used, which is an adaptive method that regulates the learning rate based on gradients. The loss function chosen is categorical crossentropy, which is suitable for two-class classification problems. Accuracy is chosen as the metric to evaluate the model performance.

To avoid overfitting, two key methods are used: EarlyStopping, which stops training if there is no improvement on the validation data within 5 epochs, and ReduceLROnPlateau, which reduces the

learning rate by 50% if the accuracy on the validation data does not improve within 5 epochs. This helps the model adapt to the data more smoothly during the training phase and avoid overfitting to the training data.

The model was trained on 50 epochs, but Early Stopping finished training on 26.

Below is the confusion matrix (fig. 4) built from the model training results. As can be seen from it, the model performs well overall, with an accuracy of 94%. However, the model is slightly less effective in recognizing the non-N class, as can be seen from the lower recall value (0.89) and the higher number of errors in the confusion matrix (223 incorrect predictions for the non-N class). However, the precision for the non-N class is very high (0.96).

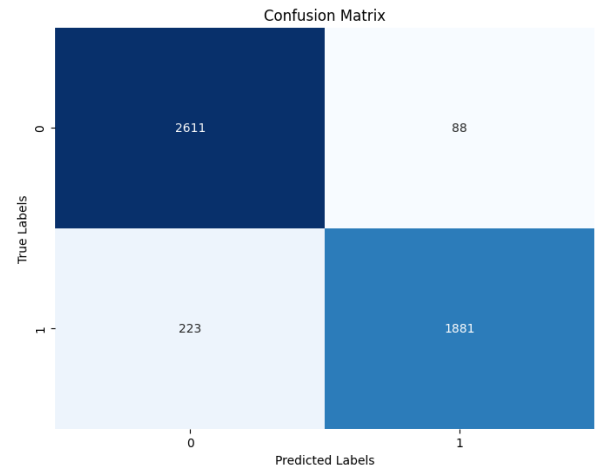


Figure 4. Confusion matrix of CNN model for N vs non-N classification.

Model for N, S, V classification

The task of dividing pico into three classes is naturally more complex. During experiments with different types of models, it was found that the S and V classes are generally divided quite poorly. Changes in the preprocessing steps (other filters, normalization, another approach to segmentation) did not improve the accuracy of the definition, so the same data processing process was used as for the two-class model.

To perform the classification, several types of architectures were tested: CNN, LSTM, Inception Module, VGG and ResNet. LSTM and ResNet turned out to be the most effective, and most of the work was done with them.

The final model was the ResNet model, which showed the smallest differences in the loss graph and the highest accuracy.

The model is a modification of the ResNet network for the task of signal classification. At the beginning, a convolutional layer with 64 filters and a kernel of size 15 is used, which precedes the blocks with residual connections. Each residual block consists of two Conv1D layers with a kernel of 7 and several filters that doubles as the network gets deeper.

The model includes two types of blocks: identical and convolutional. Identical blocks have an input signal that passes through two convolutional layers and is added to the original signal to form a residual connection. In convolutional blocks, a convolutional layer with reduced dimensionality is also used for the residual connection. To reduce overfitting, a fairly large value for the Dropout layers was used early 0.5. The final layer is a fully connected one with a softmax activation function, which is responsible for classification into three classes.

To prevent overfitting, EarlyStopping, ReduceLROnPlateau and class weights were used.

To improve the separation of the S and V peaks, the ability to manually adjust class weights was added, as well as new metrics to improve the resolution of class V.

To balance the classes, class weights were added to the loss function to increase the penalties for errors in classes S and V. A penalty for one of the classes led to a strong imbalance with the second class.

F1-score was added to understand and possibly improve the performance of the model with class V.

Precision/Recall for the third class calculates a weighted average between precision and recall for the third class, considering the β parameter.

A callback executed at the end of each training epoch calculates and outputs custom metrics for the third class (F1-score and Precision/Recall). This allows you to track how these metrics change as the model learns and helps understand how well the model copes with the third class.

Unfortunately, the separation of classes S and V remains quite low. When trying to balance one of the classes, the accuracy of the second dropped significantly. In fact, this is exactly what can be seen in the confusion matrix (Fig. 5).

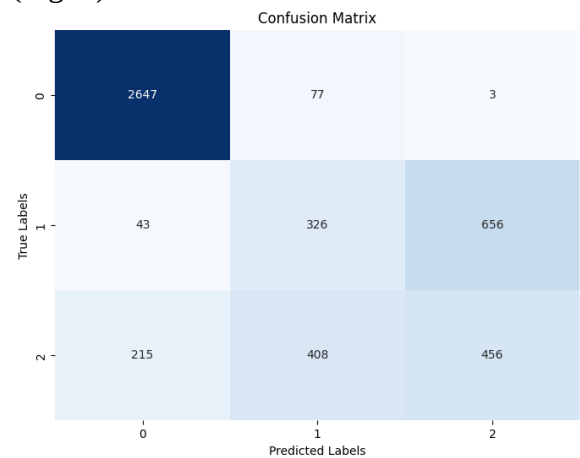


Figure 5. Confusion matrix for NSV model

As can be seen from the matrix, class N is determined well (91% accuracy), but the accuracy of determining S and V is 31 and 42%, respectively.

It can be assumed that the reason for such low accuracy is inappropriate data segmentation or an error in preprocessing, but unfortunately it was not possible to find and fix it.

References

- 1) Photoplethysmography (PPG)
[https://www.news-medical.net/health/Photoplethysmography-\(PPG\).](https://www.news-medical.net/health/Photoplethysmography-(PPG).)
- 2) Elisa Mejía-Mejía, John Allen, Karthik Budidha, Chadi El-Hajj, Panayiotis A. Kyriacou and Peter H. Charlton. Photoplethysmography Signal Processing and Synthesis
- 3) Liu, S.-H.; Li, R.-X.; Wang, J.-J.; Chen, W.; Su, C.-H. Classification of Photoplethysmographic Signal Quality with Deep Convolution Neural Networks for Accurate Measurement of Cardiac Stroke Volume. *Appl. Sci.* 2020, 10, 4612.
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